Supplementary Information

Delta Dependency Prioritizes Paralog-Based
Synthetic Lethality Candidates Across Solid Tumor Types

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1 Supplementary Figures

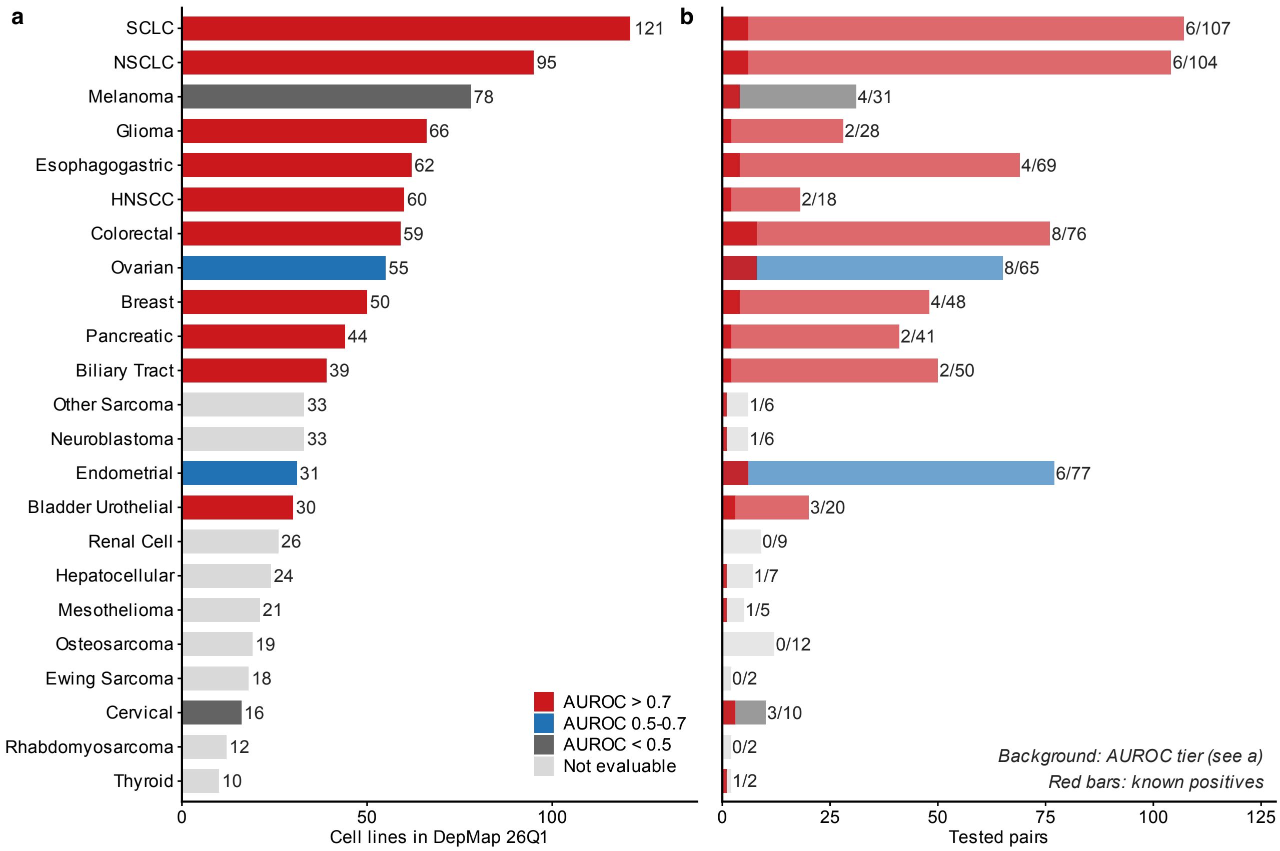

Fig. S1: Cell line landscape and evaluation summary across 23 solid tumor types. a, Cell line counts per cancer type, colored by DD AUROC tier (red: AUROC > 0.7; blue: 0.5–0.7; gray: < 0.5; light gray: not evaluable). b, Tested paralog pairs per cancer type; red bars show known gold-standard positives. Numbers indicate known/total pairs.

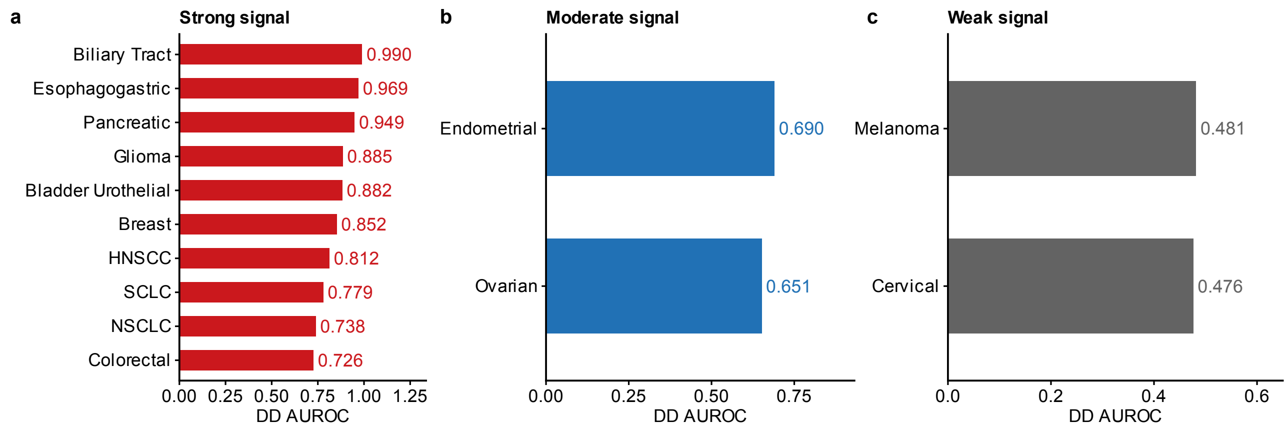

Fig. S2: Cross-cancer DD AUROC. DD AUROC for each of the 14 evaluable solid tumor types, grouped by signal strength. a, Strong signal (AUROC $\geq$ 0.7; 10 lineages including Biliary Tract 0.990, Esophagogastric 0.969, Pancreatic 0.949). b, Moderate signal (AUROC 0.5–0.7; Endometrial 0.690, Ovarian 0.651). c, Weak signal (AUROC < 0.5; Melanoma 0.481, Cervical 0.476). Dashed line indicates random classifier (0.5). Bar labels show the number of tested paralog pairs per lineage. Cancer types with <2 known positive pairs were excluded from AUROC evaluation (see Table S1 for the 9 non-evaluable types).

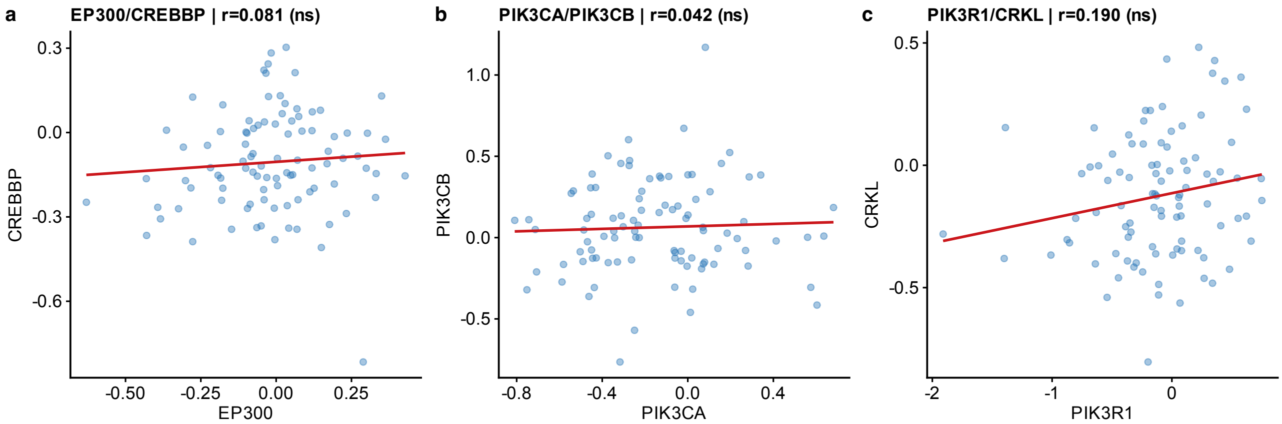

Fig. S3: CPTAC per-cohort scatter plots (UCEC). Representative scatter plot of EP300 vs. CREBBP protein abundance ($\log_2$ normalized intensity) in the UCEC cohort ($n = 83$). Each point represents one tumor sample. Line: linear regression fit with 95% confidence band. Pearson r and BH-corrected q-value shown. Similar scatter plots for other cohorts and paralog pairs are available upon request.

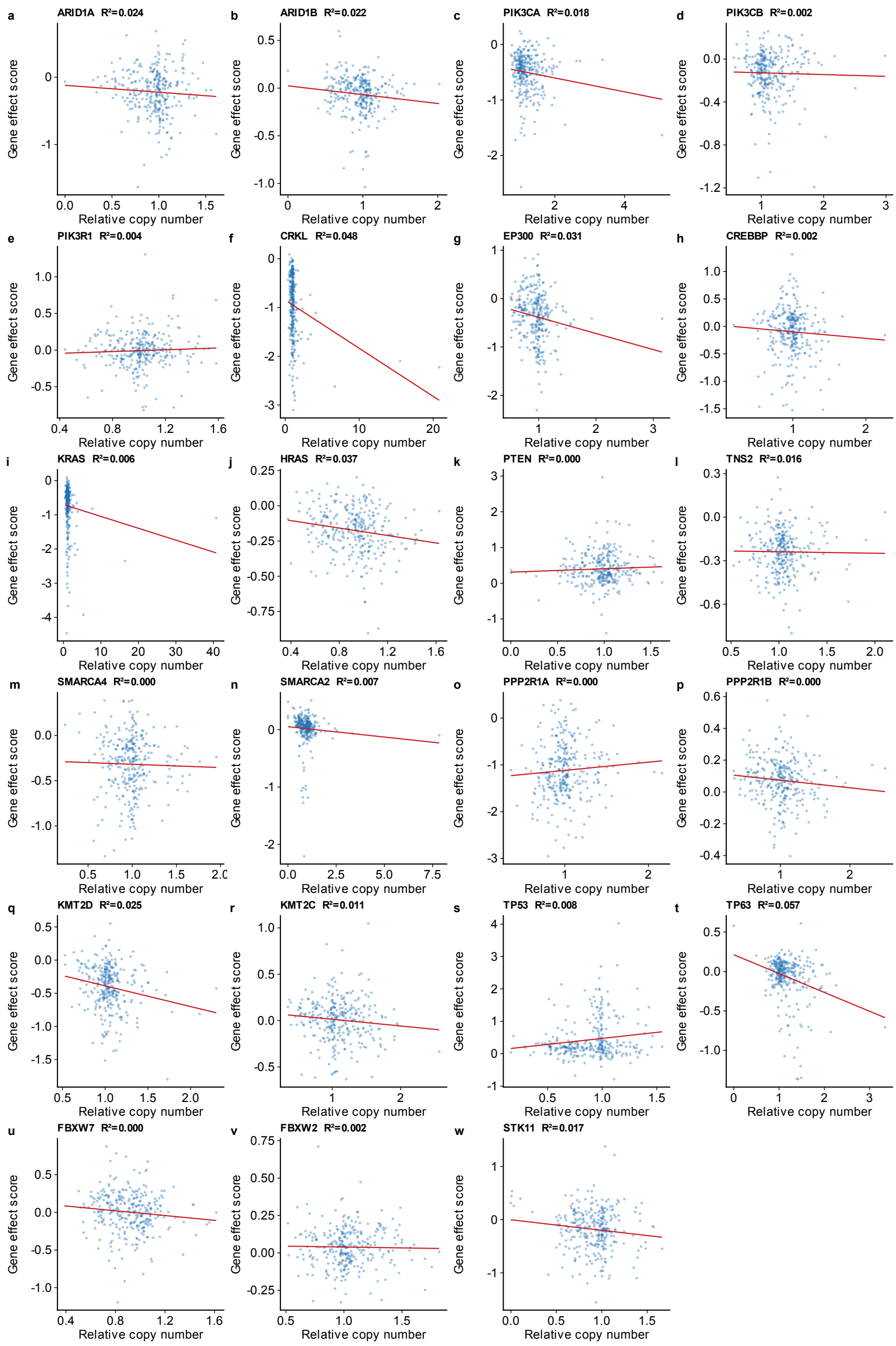

Fig. S4: CNV independence analysis. Scatter plots of relative copy number (x-axis) vs. Chronos gene-effect score (y-axis) for each of 23 paralog genes across 1,192 cell lines with both CNV and dependency data. Each panel shows one gene with the linear regression fit (red line) and R^2 value. All $R^2 < 0.10$, indicating that copy number variation explains <10% of gene-effect variance. Multivariate regression controlling for driver mutation and TP53 co-mutation left DD estimates essentially unchanged after CNV adjustment (see main text).

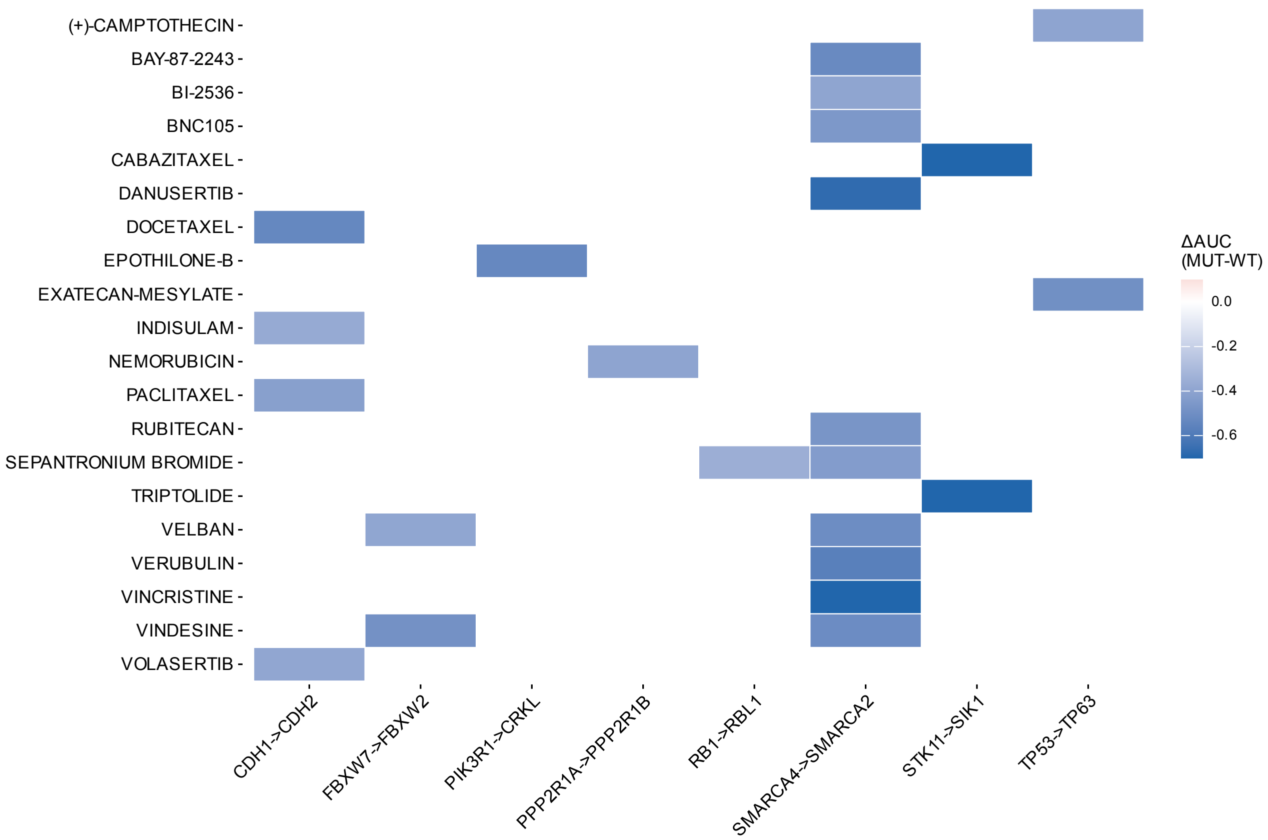

Fig. S5: PRISM drug selectivity heatmap. Heatmap of drug sensitivity differential (ΔAUC = mean AUC in driver-mutant minus mean AUC in wild-type cell lines) for the top compound–paralog pair associations from the PRISM Repurposing screen (1,482 compounds, 727 cell lines). Negative ΔAUC indicates selective sensitivity in driver-mutant cells. Associations shown met discovery-stage criteria (BH $q < 0.25$, $\Delta\text{AUC} < 0$, enrichment> 0.1). Known drug–target biology is recovered: MEK inhibitors with KRAS-mutant lines, mTOR/AKT inhibitors with PTEN-mutant lines, HDAC inhibitors with EP300-mutant lines.

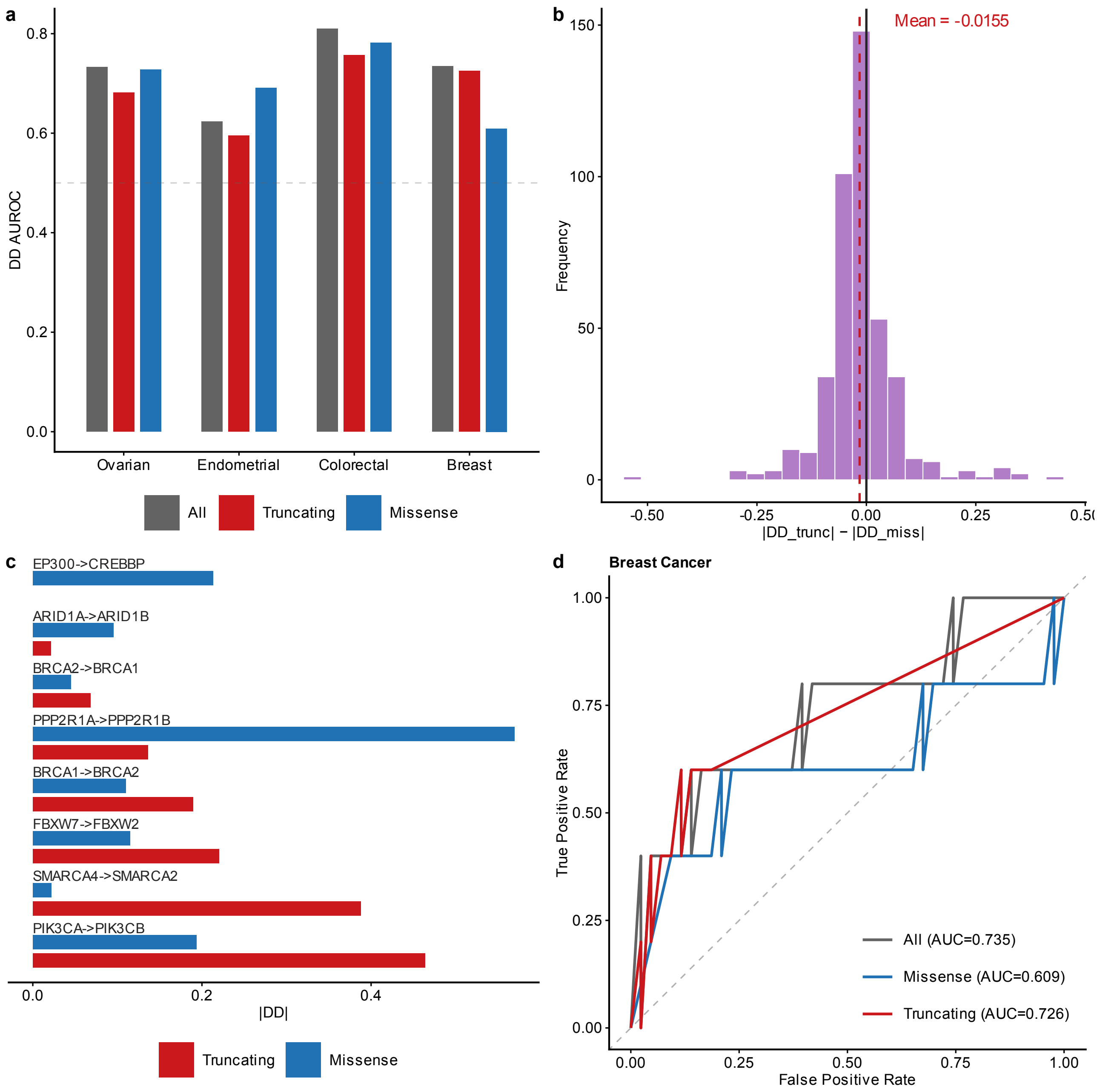

Fig. S6: Mutation-type stratification. a, DD AUROC stratified by mutation type (truncating vs. missense) across cancer types. b, Distribution of ΔDD (truncating minus missense) across all tested driver–paralog pairs; overall mean near zero (−0.005), indicating that truncating > missense is pair-specific rather than universal. c, ΔDD for known gold-standard SL pairs showing that ARID1A→ARID1B (+0.368 in Ovarian) and EP300→CREBBP (+0.314 in Colorectal) drive the trend. d, ROC curves for Breast cancer comparing truncating-only (AUROC = 0.726) vs. missense-only (0.391) and overall DD performance. Panel d is computed on the mutation-type analysis frame (48 Breast pairs); the resulting "All" AUC (0.735) therefore differs from the cross-cancer frame value (0.852; Fig. 1a), which uses a different pair universe.

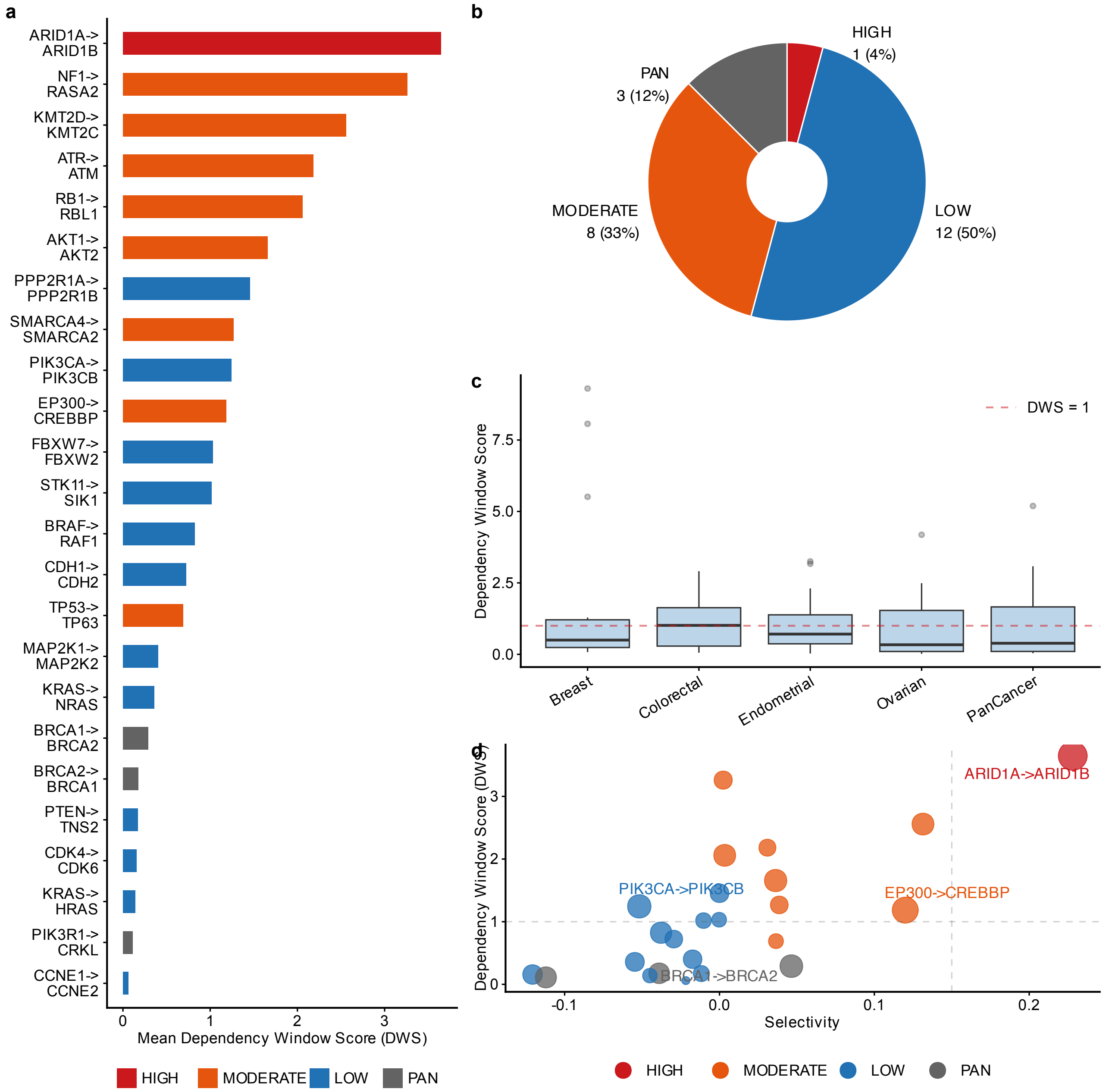

Fig. S7: Therapeutic window analysis. a, All 24 paralog pairs ranked by mean dependency window score (DWS) across 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer). ARID1A→ARID1B ranks first (DWS = 3.65). b, In vitro selectivity tier classification: HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0), MODERATE, LOW_SELECTIVITY, and PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$). c, DWS values broken down by individual cancer context showing cross-lineage consistency for top candidates. d, Selectivity vs. DWS scatter plot; ARID1A→ARID1B is the only pair in the upper-right quadrant (high DWS and high selectivity).

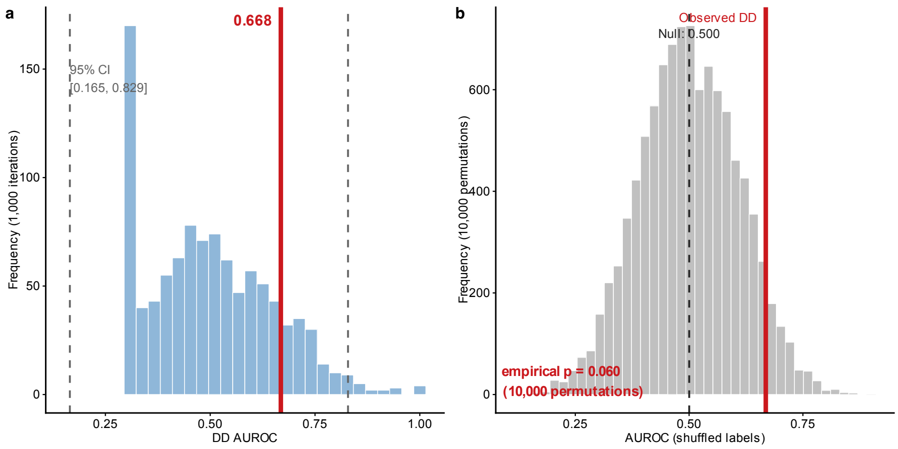

Fig. S8: Bootstrap and negative control (per-pair evaluation, 75 pairs, 6 known positives). a, Bootstrap distribution (1,000 iterations; 95% CI: 0.165–0.829). b, Null distribution from 10,000 label-shuffled permutations (empirical p = 0.530). The observed per-pair AUROC (0.493) does not exceed the null mean (0.501): with only 6 positive pairs the aggregated per-pair evaluation has no detectable signal, and the lineage-level evaluation (0.682, reported in the benchmark comparison) serves as the primary framework because it evaluates each driver×paralog×lineage combination separately rather than aggregating across lineages.

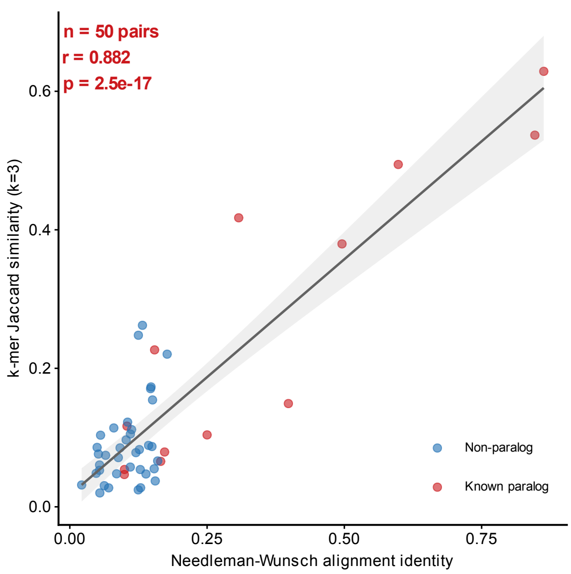

Fig. S9: k-mer Jaccard vs. Needleman-Wunsch alignment identity validation. Scatter plot of k-mer Jaccard similarity ($k = 3$) vs. global alignment identity for 50 protein pairs (13 known paralogs, 37 non-paralog controls). Red: known paralog-SL pairs; Blue: non-paralog pairs. Gray line: linear regression with 95% CI.

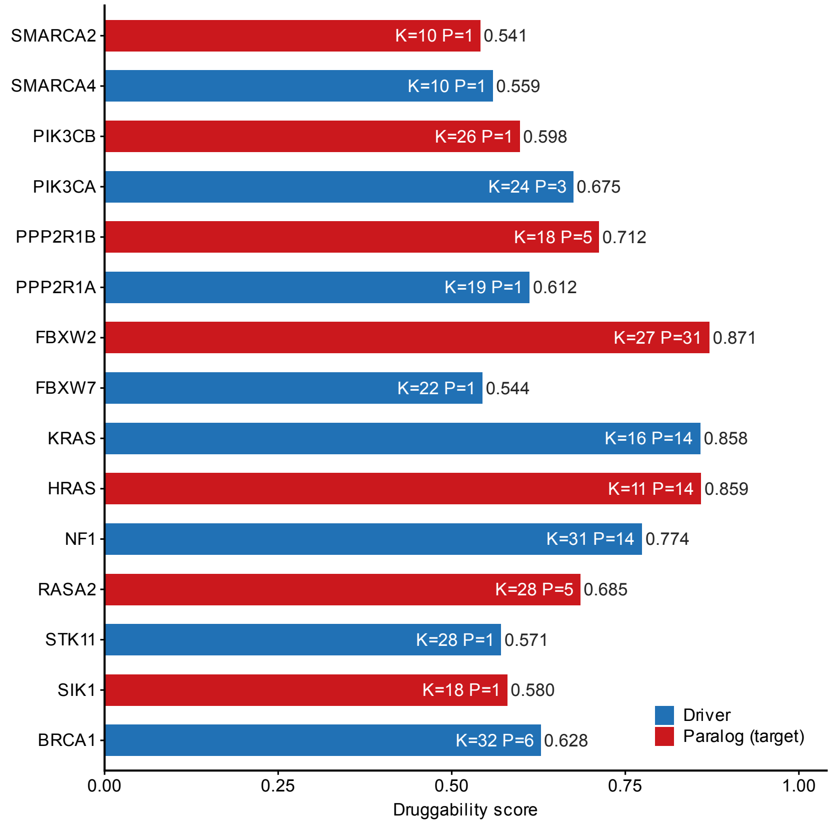

Fig. S10: Structure-based druggability assessment. Composite druggability score for 15 proteins (7 paralog targets in red, 8 driver genes in blue) based on ESMFold-predicted structures. Scores integrate structural confidence (pLDDT), hydrophobic content, lysine density (K, relevant for PROTAC conjugation), and predicted pocket count (P). Structures were limited to 400 residues per protein; ARID1A/ARID1B structures were not available from AlphaFold DB and are assessed by sequence-level features only (main text Table 2). FBXW2 (0.871) and KRAS/HRAS (0.858–0.859) show the highest scores due to well-folded compact domains.

2 Supplementary Tables

2.1 Table S1: Cell line and evaluation summary across 23 solid tumor types

Label	Full name	Cell Lines	Pairs	Known+	DD AUROC
Biliary Tract	Biliary tract cancer	39	50	2	0.990
Esophagogastric	Esophagogastric adenocarcinoma / esophageal SCC	62	69	4	0.969
Pancreatic	Pancreatic adenocarcinoma	44	41	2	0.949
Glioma	Diffuse glioma	66	28	2	0.885
Bladder Urothelial	Bladder urothelial carcinoma	30	20	3	0.882
Breast	Invasive breast carcinoma	50	48	4	0.852
HNSCC	Head and neck squamous cell carcinoma	60	18	2	0.812
SCLC	Small cell lung cancer	121	107	6	0.779
NSCLC	Non-small cell lung cancer	95	104	6	0.738
Colorectal	Colorectal adenocarcinoma	59	76	8	0.726
Endometrial	Endometrial carcinoma	31	77	6	0.690
Ovarian	Ovarian epithelial carcinoma	55	65	8	0.651
Melanoma	Melanoma	78	31	4	0.481
Cervical	Cervical carcinoma	16	10	3	0.476
Mesothelioma	Pleural mesothelioma	21	5	1	—
Hepatocellular	Hepatocellular carcinoma	24	7	1	—
Renal Cell	Renal cell carcinoma	26	9	0	—
Neuroblastoma	Neuroblastoma	33	6	1	—
Osteosarcoma	Osteosarcoma	19	12	0	—
Ewing Sarcoma	Ewing sarcoma	18	2	0	—
Rhabdomyosarcoma	Rhabdomyosarcoma	12	2	0	—
Other Sarcoma	Other soft-tissue sarcoma	33	6	1	—
Thyroid	Thyroid carcinoma	10	2	1	—

Table S1: Cell line counts and DD AUROC across all 23 analyzed solid tumor types. Top: AUROC > 0.7 (10 types). Middle: AUROC ≤ 0.7 (4 types). Bottom: insufficient known positives for AUROC (9 types). Known+: number of gold-standard positive pairs evaluable in each lineage. "—": AUROC not computable (<2 known positives). Full name: standardized cancer-type name corresponding to each short label; SCC, squamous cell carcinoma.

2.2 Table S2: Complete paralog-SL pair analysis results

The full analysis results are provided as a tab-separated file: `TableS2_FullResults.tsv` (116 rows $\times$ 13 columns, representing all driver$\times$paralog$\times$cancer-type combinations passing the minimum sample-size filter).

Data dictionary:

- `driver_gene`: Driver gene with damaging mutation.
- `paralog_gene`: Candidate paralog tested for conditional dependency.
- `cancer_type`: Solid tumor lineage (DepMap OncotreePrimaryDisease).
- `pcs`: Paralog Compensation Score (DD $\times$ necessity).
- `delta_expression`: ΔExpression between MUT and WT groups.
- `necessity`: Mean gene-effect score of the paralog (more negative = more essential).
- `dependency_dd`: Delta Dependency (signed; positive = greater dependency in MUT).
- `composite_score`: Weighted composite of PCS, DD, and ΔExpression.
- `novelty`: "Known" if in gold-standard set (Table S3), else "Novel".

- `q_value`: Benjamini-Hochberg adjusted p-value within each driver gene.
- `mutation_frequency`: Fraction of cell lines with driver mutation in this lineage.
- `n_mut`, `n_wt`: Number of mutant and wild-type cell lines.
- `is_known_paralog_sl`: Boolean; TRUE if pair is in gold-standard set.

2.3 Table S3: Gold-standard paralog-SL pairs with evidence provenance and tier assignment

Tier	Driver	Paralog	Type	Evidence	Key Ref	Indep.
A	SMARCA4	SMARCA2	Seq. paralog	CRISPR screen	Hoffman 2014	Yes
A	ARID1A	ARID1B	Seq. paralog	CRISPR + drug	Helming 2014	Yes
A*	EP300	CREBBP	Seq. paralog	Reciprocal SL (CREBBP→EP300)	Ogiwara 2016; Nie 2021	Yes
comparator	BRCA1	BRCA2	Func. analog	Genetic screen	Bryant 2005	Yes
comparator	STK11	SIK1	Partial hom.	LKB1–SIK axis (mouse genetics)	Hollstein 2019	Yes
B	AKT1	AKT2	Seq. paralog	Digenic KO (comb. CRISPR)	Najm 2018	Yes
B	CCNE1	CCNE2	Seq. paralog	Mouse double-KO redundancy	Geng 2003	Yes
B	PIK3CA	PIK3CB	Seq. paralog	Redundancy/pharmacologic	Wee 2008 (PTEN→PIK3CB only)	Yes
B	CDK4	CDK6	Seq. paralog	Digenic KO (pgPEN) + drug	Parrish 2021	Yes
B	MAP2K1	MAP2K2	Seq. paralog	Digenic KO (pgPEN) + drug	Parrish 2021	Yes
C	FBXW7	FBXW2	Seq. paralog	DepMap analysis	DepMap	No
C	PPP2R1A	PPP2R1B	Seq. paralog	DepMap analysis	DepMap	No

Table S3: Twelve curated pairs with evidence provenance and tier assignment. Tier A (2 pairs): directional external experimental evidence matching the direction scored here; used for directional claims. A* (1 pair): synthetic lethality experimentally established in the reciprocal direction only (CREBBP→EP300); the direction scored here (EP300→CREBBP) is supported by paralog redundancy alone, so the pair is excluded from directional Tier A claims and relabelled non-positive in the direction-strict sensitivity analysis. Comparator (2 pairs): mechanistic reference pairs (functional analog or pathway axis), reported separately. Tier B (5 pairs): paralog-redundancy, digenic-knockout, or pharmacologic evidence only, not directional genotype-conditional. Tier C (2 pairs): DepMap-derived, excluded from directional claims to avoid circularity. Type: Seq. paralog (shared gene family, conserved domains), Func. analog (shared pathway role, no sequence homology), Partial hom. (same superfamily, divergent). Indep.: whether evidence is independent of DepMap CRISPR data. "No" indicates pairs whose SL status was originally identified from DepMap-derived analyses, creating potential circularity.

2.4 Table S4: Pan-cancer summary statistics

Cancer Type	Cell Lines	Pairs	DD AUROC
Biliary Tract	39	50	0.990
Esophagogastric	62	69	0.969
Pancreatic	44	41	0.949
Glioma	66	28	0.885
Bladder Urothelial	30	20	0.882
Breast	50	48	0.852
HNSCC	60	18	0.812
SCLC	121	107	0.779
NSCLC	95	104	0.738
Colorectal	59	76	0.726
Endometrial	31	77	0.690
Ovarian	55	65	0.651
Melanoma	78	31	0.481
Cervical	16	10	0.476

Table S4: DD AUROC across all 14 evaluable solid tumor types (sorted by AUROC). Horizontal line separates AUROC > 0.7 (top, 10 lineages) from AUROC ≤ 0.7 (bottom). Full lineage names are given in Table S1.

2.5 Table S5: 40-gene driver panel

Gene	Cancer Types	Source	Category
TP53	Pan-cancer	TCGA+COSMIC	TSG
KRAS	Lung, CRC, Pancreatic	TCGA+COSMIC	Oncogene
PIK3CA	Breast, Endometrial, Cervical	TCGA+COSMIC	Oncogene
PTEN	Endometrial, Ovarian, Breast	TCGA+COSMIC	TSG
ARID1A	Endometrial, Ovarian, CRC	TCGA+COSMIC	TSG
BRCA1	Ovarian, Breast	TCGA+COSMIC	TSG
BRCA2	Ovarian, Breast	TCGA+COSMIC	TSG
BRAF	Melanoma, CRC, Thyroid	TCGA+COSMIC	Oncogene
NF1	Glioma, Breast, Lung	TCGA+COSMIC	TSG
RB1	SCLC, Breast, Bladder	TCGA+COSMIC	TSG

Table S5: First 10 of 40 driver genes (complete list in TableS5_DriverGenes.tsv). Selection criteria: recurrently mutated in ≥1 TCGA pan-cancer study or COSMIC Cancer Gene Census, with ≥3 mutant cell lines in at least one DepMap lineage.

2.6 Table S6: All 24 paralog pairs ranked by preclinical prioritization score

Driver	Paralog	|DD|	DWS	Select.	Class	Type
ARID1A	ARID1B	0.250	3.647	0.228	HS	Seq
NF1	RASA2	0.062	3.260	0.002	MOD	Seq
KMT2D	KMT2C	0.112	2.558	0.131	MOD	Seq
ATR	ATM	0.049	2.182	0.031	MOD	Seq
RB1	RBL1	0.112	2.059	0.003	MOD	Seq
AKT1	AKT2	0.119	1.657	0.036	MOD	Seq
PPP2R1A	PPP2R1B	0.068	1.454	0.000	LS	Seq
SMARCA4	SMARCA2	0.057	1.268	0.039	MOD	Seq
PIK3CA	PIK3CB	0.137	1.245	−0.052	LS	Seq
EP300	CREBBP	0.187	1.185	0.120	MOD	Seq
FBXW7	FBXW2	0.030	1.032	0.000	LS	Seq
STK11	SIK1	0.037	1.018	−0.010	LS	Part
BRAF	RAF1	0.105	0.824	−0.038	LS	Seq
CDH1	CDH2	0.057	0.722	−0.030	LS	Seq
TP53	TP63	0.031	0.690	0.036	MOD	Seq
MAP2K1	MAP2K2	0.067	0.401	−0.017	LS	Seq
KRAS	NRAS	0.071	0.358	−0.055	LS	Seq
BRCA1	BRCA2	0.125	0.290	0.046	PE	Func
BRCA2	BRCA1	0.094	0.177	−0.039	PE	Func
PTEN	TNS2	0.041	0.172	−0.012	LS	Seq
CDK4	CDK6	0.079	0.157	−0.121	LS	Seq
KRAS	HRAS	0.028	0.139	−0.045	LS	Seq
PIK3R1	CRKL	0.100	0.111	−0.112	PE	Seq
CCNE1	CCNE2	0.004	0.060	−0.022	LS	Seq

Table S6: All 24 analyzed paralog pairs ranked by mean dependency window score (DWS). |DD|: mean absolute Delta Dependency across cancer contexts. Select.: mean selectivity (fraction mutant-essential minus fraction wild-type-essential). Class: HS=HIGH_SELECTIVITY, MOD=MODERATE, LS=LOW_SELECTIVITY, PE=PAN_ESSENTIAL. Type: Seq=sequence paralog, Part=partial homology, Func=functional analog.

2.7 Table S7: Gene-class-specific driver-mutation rules and variant classification

Gene	Class	Rule	Old	New	% ret.
ALK	ONC	Hotspot	118	12	10.2
APC	TSG	LikelyLoF	217	108	49.8
ARID1A	TSG	LikelyLoF	240	173	72.1
ATM	TSG	LikelyLoF	165	59	35.8
ATR	TSG	LikelyLoF	125	37	29.6
BRAF	ONC	Hotspot	212	162	76.4
BRCA1	TSG	LikelyLoF	101	34	33.7
BRCA2	TSG	LikelyLoF	159	68	42.8
CCNE1	ONC	Hotspot	22	0	0.0
CDH1	TSG	LikelyLoF	87	34	39.1
CDK4	ONC	Hotspot	27	8	29.6
CTNNB1	ONC	Hotspot	85	45	52.9
EGFR	ONC	Hotspot	110	19	17.3
EP300	TSG	LikelyLoF	206	107	51.9
ERBB2	ONC	Hotspot	70	20	28.6
FBXW7	TSG	LikelyLoF	116	81	69.8
GATA3	TSG	LikelyLoF	52	16	30.8
KEAP1	TSG	LikelyLoF	99	31	31.3
KMT2D	TSG	LikelyLoF	356	185	52.0
KRAS	ONC	Hotspot	254	233	91.7
MAP2K1	ONC	Hotspot	36	14	38.9
MAP3K1	TSG	LikelyLoF	74	14	18.9
MAPK1	ONC	Hotspot	13	0	0.0
MET	ONC	Hotspot	66	1	1.5
NF1	TSG	LikelyLoF	215	122	56.7
PIK3CA	ONC	Hotspot	182	153	84.1
PIK3R1	TSG	LikelyLoF	79	46	58.2
PPP2R1A	TSG	LikelyLoF	53	13	24.5
PTEN	TSG	LikelyLoF	199	176	88.4
RB1	TSG	LikelyLoF	185	150	81.1
SMAD4	TSG	LikelyLoF	119	80	67.2
STK11	TSG	LikelyLoF	97	66	68.0
TP53	TSG	LikelyLoF	1171	1140	97.4

Table S7: Driver-mutation definitions by gene class (DepMap 26Q1, default-entry profiles). TSG: mutant requires `LikelyLoF`; ONC: mutant requires `Hotspot`. Old: cell lines carrying any non-silent variant (pre-revision permissive definition); New: cell lines qualifying under the class-specific rule; % ret.: 100 × New/Old. The top VariantInfo terms per gene are at `output/driver_mutation_rules.csv`. CCNE1 and MAPK1 are amplification-driven oncogenes with no qualifying hotspot mutations.